

Two-Stage CMOS Op-Amp with Miller Compensation

Hand Calculations & Design Derivation

Analog IC Design Portfolio

180 nm CMOS | VDD = 1.8 V

1. Specifications and process constants

The design follows the classic Allen–Holberg two-stage OTA methodology: a PMOS differential input pair with an NMOS current-mirror load, driving an NMOS common-source second stage with a PMOS current-source load, compensated by a Miller capacitor C_c and a series nulling resistor R_z .

Target	Symbol	Value
DC open-loop gain	A_{v0}	≥ 60 dB
Unity-gain bandwidth	GBW	10 MHz
Phase margin	PM	$\geq 60^\circ$
Load capacitance	C_L	10 pF
Slew rate	SR	≥ 10 V/ μ s

Process (generic 180 nm): $\mu_n C_{ox} = 270 \mu\text{A V}^{-2}$, $\mu_p C_{ox} = 70 \mu\text{A V}^{-2}$, $|V_{th}| = 0.45$ V, $\lambda = 0.08 \text{ V}^{-1}$ at $L = 0.5 \mu\text{m}$.

2. Compensation capacitor and tail current

For a phase margin of 60° the standard rule requires $C_c > 0.22 C_L$. Choosing a comfortable $0.3 C_L$:

$$C_c = 0.3 C_L = 0.3 \times 10 \text{ pF} = \boxed{3 \text{ pF}}$$

The first-stage slew rate is set by how fast the tail current I_5 can charge C_c :

$$\text{SR} = \frac{I_5}{C_c} \Rightarrow I_5 = \text{SR} \cdot C_c = (10 \times 10^6)(3 \times 10^{-12}) = \boxed{30 \mu\text{A}}$$

Each input branch therefore carries $I_1 = I_2 = I_5/2 = 15 \mu\text{A}$.

3. Input pair transconductance from GBW

The unity-gain bandwidth of a Miller-compensated two-stage amplifier is

$$\text{GBW} = \frac{g_{m1}}{C_c} \Rightarrow g_{m1} = 2\pi \text{GBW} \cdot C_c = 2\pi(10^7)(3 \times 10^{-12}) = \boxed{188.5 \mu\text{S}}$$

Sizing the PMOS input pair from $g_m = \sqrt{2\mu_p C_{ox} (W/L) I_D}$:

$$\left(\frac{W}{L}\right)_{1,2} = \frac{g_{m1}^2}{2\mu_p C_{ox} I_1} = \frac{(188.5 \times 10^{-6})^2}{2(70 \times 10^{-6})(15 \times 10^{-6})} \approx \boxed{17}$$

With $L = 0.5 \mu\text{m}$ this gives $W_{1,2} \approx 8.5 \mu\text{m}$. The corresponding overdrive is $V_{ov1} = 2I_1/g_{m1} \approx 159 \text{mV}$.

4. Mirror load, tail and bias sizing

Mirror load M3/M4 (NMOS) — size for $V_{ov3} = 0.2 \text{V}$:

$$\left(\frac{W}{L}\right)_{3,4} = \frac{2I_1}{\mu_n C_{ox} V_{ov3}^2} = \frac{2(15 \times 10^{-6})}{(270 \times 10^{-6})(0.2)^2} \approx \boxed{3}, \quad g_{m3} = \frac{2I_1}{V_{ov3}} = 150 \mu\text{S}$$

Tail source M5 (PMOS) — size for $V_{ov5} = 0.3 \text{V}$:

$$\left(\frac{W}{L}\right)_5 = \frac{2I_5}{\mu_p C_{ox} V_{ov5}^2} = \frac{2(30 \times 10^{-6})}{(70 \times 10^{-6})(0.3)^2} \approx \boxed{10}$$

Bias diode M8 mirrors 1:1 from a $30 \mu\text{A}$ reference, so $(W/L)_8 = (W/L)_5 = 10$.

5. Second stage sizing for phase margin

To place the non-dominant pole at $p_2 = 2.2 \cdot \omega_{\text{GBW}}$ (the condition for $\approx 60^\circ$ PM), the second-stage transconductance must be

$$g_{m6} = 2.2 \omega_{\text{GBW}} C_L = 2.2 (2\pi \cdot 10^7)(10 \times 10^{-12}) \approx \boxed{1.38 \text{mS}}$$

For **zero systematic offset**, M6 shares its gate–source voltage with the mirror (M3/M4), so $V_{ov6} = V_{ov3} = 0.2 \text{V}$ and

$$I_6 = \frac{1}{2} g_{m6} V_{ov6} = \frac{1}{2} (1.38 \times 10^{-3})(0.2) \approx \boxed{138 \mu\text{A}}$$

$$\left(\frac{W}{L}\right)_6 = \left(\frac{W}{L}\right)_3 \frac{I_6}{I_3} = 3 \cdot \frac{138}{15} \approx \boxed{26} \quad (W_6 \approx 13 \mu\text{m})$$

The PMOS load M7 carries the same $I_7 = I_6 = 138 \mu\text{A}$. Zero-offset mirroring requires $(W/L)_7/(W/L)_5 = I_7/I_5$:

$$\left(\frac{W}{L}\right)_7 = \left(\frac{W}{L}\right)_5 \frac{I_7}{I_5} = 10 \cdot \frac{138}{30} \approx \boxed{44} \quad (W_7 \approx 22 \mu\text{m})$$

6. Open-loop gain

Output resistances (using $r_o = 1/(\lambda I_D)$):

$$R_{o1} = r_{o2} \parallel r_{o4} = \frac{1}{2\lambda I_1} = \frac{1}{2(0.08)(15 \mu\text{A})} \approx 417 \text{ k}\Omega$$

$$R_{o2} = r_{o6} \parallel r_{o7} = \frac{1}{2\lambda I_6} = \frac{1}{2(0.08)(138 \mu\text{A})} \approx 45.3 \text{ k}\Omega$$

Stage gains:

$$A_{v1} = g_{m1} R_{o1} = (188.5 \mu\text{S})(417 \text{ k}\Omega) \approx 78.6 \text{ (37.9 dB)}$$

$$A_{v2} = g_{m6} R_{o2} = (1.38 \text{ mS})(45.3 \text{ k}\Omega) \approx 62.6 \text{ (35.9 dB)}$$

$$\boxed{A_{v0} = A_{v1} A_{v2} \approx 4920 \approx 73.8 \text{ dB}} \geq 60 \text{ dB } \checkmark$$

7. Pole splitting — the Miller effect

Without compensation the two high-impedance nodes sit at similar frequencies, giving a poor phase margin. Connecting C_c between the input and output of the inverting second stage makes it appear $(1 + g_{m6} R_{o2})$ times larger at the first-stage node (the **Miller multiplication**), pushing the dominant pole *down* while pushing the second pole *up*:

$$\omega_{p1} \approx \frac{1}{R_{o1} g_{m6} R_{o2} C_c} = \frac{1}{(417\text{k})(1.38\text{m})(45.3\text{k})(3\text{p})} \approx 1.28 \times 10^4 \text{ rad/s}$$

$$\boxed{f_{p1} \approx 2.0 \text{ kHz}}$$

$$\omega_{p2} \approx \frac{g_{m6}}{C_L} = \frac{1.38 \times 10^{-3}}{10 \times 10^{-12}} = 1.38 \times 10^8 \text{ rad/s} \Rightarrow \boxed{f_{p2} \approx 22 \text{ MHz}}$$

Sanity check on bandwidth: $\text{GBW} = A_{v0} f_{p1} \approx 4920 \times 2.0 \text{ kHz} \approx 10 \text{ MHz } \checkmark$ (equivalently g_{m1}/C_c).

8. The right-half-plane zero and the nulling resistor

The Miller capacitor also creates a feed-forward path that produces a **right-half-plane (RHP) zero**:

$$\omega_z = \frac{g_{m6}}{C_c} = \frac{1.38 \times 10^{-3}}{3 \times 10^{-12}} = 4.6 \times 10^8 \text{ rad/s} \Rightarrow f_z \approx 73 \text{ MHz}$$

A RHP zero boosts magnitude but subtracts phase — it *degrades* stability. The fix is a **series nulling resistor** R_z , which relocates the zero to

$$\omega_z = \frac{1}{C_c \left(\frac{1}{g_{m6}} - R_z \right)}.$$

Two useful design points:

- $R_z = 1/g_{m6} \approx 725 \Omega$ pushes the zero to infinity.
- $R_z = \frac{1}{g_{m6}} \left(1 + \frac{C_L}{C_c} \right) = 725 (1 + 10/3) \approx \boxed{3.1 \text{ k}\Omega}$ places a **left-half-plane** zero exactly on p_2 , cancelling it.

9. Phase margin

With the Miller cap alone (RHP zero present):

$$\text{PM} = 180^\circ - \tan^{-1} \frac{\text{GBW}}{f_{p1}} - \tan^{-1} \frac{\text{GBW}}{f_{p2}} - \tan^{-1} \frac{\text{GBW}}{f_z} \approx 58^\circ$$

After adding R_z (RHP zero removed, p_2 cancelled), the phase lag near the crossover collapses and

$$\boxed{\text{PM} \approx 66^\circ\text{--}81^\circ} \geq 60^\circ \checkmark$$

10. Slew rate and power

$$\text{SR} = \frac{I_5}{C_c} = \frac{30 \mu\text{A}}{3 \text{ pF}} = \boxed{10 \text{ V } \mu\text{s}^{-1}} \quad (\text{second stage: } I_7/C_L = 13.8 \text{ V } \mu\text{s}^{-1} > \text{SR } \checkmark)$$

$$P = V_{DD}(I_{\text{ref}} + I_5 + I_7) = 1.8 (30 + 30 + 138) \mu\text{A} \approx \boxed{357 \mu\text{W}}$$

11. Summary

Quantity	Symbol	Result
DC gain	A_{v0}	73.8 dB
Unity-gain BW	GBW	10.0 MHz
Dominant pole	f_{p1}	2.0 kHz

Quantity	Symbol	Result
Non-dominant pole	f_{p2}	22 MHz
RHP zero (no R_z)	f_z	73 MHz
Miller cap	C_c	3 pF
Nulling resistor	R_z	3.1 k Ω
Phase margin (final)	PM	$\geq 66^\circ$
Slew rate	SR	10 V/ μ s
Power	P	0.36 mW

These hand-calculated targets are the reference against which the Cadence Spectre simulations (see `../results/`) are verified.